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Androids take over the world!

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COMP 5130

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12/5/2022

Abstract

The primary focus of this paper will be to examine how designing mobile apps in the past was difficult before the introduction of android studio IDE and how it has made work easier with its advanced high-end features. As a compiler nowadays used in designing most apps it is important to understand the platform better and why it is useful in today's android programming as opposed to before.

Android studio is a relatively new IDE (Integrated Development Environment) that allows its users to create android applications of all different types of complexity. With android studio, programmers are in a better position to make good use of the in-built features that make building android applications a bit easier, more robust, and more interesting. The IDE platform comes with a jetpack compose tool, to enhance building a nice native user interface (UI). The toolkit comes in handy by enabling a software developer to use fewer codes, intuitive, and very powerful in building some of the best apps.

Another important feature that comes as an inbuilt utility with the IDE is the intelligent code editor. The intelligent code editor makes it easier for developers to be able witnesses any changes in their codes. Besides, it gives the freedom of writing better codes having been automated with an internal intelligent tool for providing users autocompletion of codes in Java language, Kotlin programming, C/C++, and any other programming languages. Furthermore, the android studio has also other significant features that help in building a successful project for any version of the device and all through the testing of the project to its deployment. Thanks to both Gradle and the use of an emulator to test various layouts that make it possible for any android app to fit any android device.

Introduction

Before diving into the paper's content, it is wise to check back on the history of android and how it made the work of building apps a bit easier. Dating back to 2010, a dedicated team in google company worked tirelessly with the end goal to introduce the android Operating System (OS) to the market that was getting crowded with smartphones. The intention was for android OS developers to develop a system whereas the manufacture focus on the hardware components of the phones. Android at that allowed developers to use the platform and build software that integrates into all kinds of devices using the same single version of the app rather than having different versions to suit various types of devices.

During such a time no one expected that android would see the success it has gained over the years. Not because it had backing from the google company but because google had a lot of ongoing projects too which if that was the case then it would have hindered the success of the android studio. Not forgetting at the same time there were a lot of competitors who had entered the market following the launch of the apple smartphone in 2007. Others that were in the market to offer options to their customers were Nokia phones, Palm phones, and Blackberry phones.

Working with a dedicated team of engineers with the end goal to deliver android version 1.0 it was clear what everyone's goal was. It was a mixture of feelings working with a team where everyone was able to raise what was right or wrong as sourced (Haase). Over the years android overcame the so-called challenges and after the struggle of the dedicated team that wanted to see it be where it is today. That begs the question of how it managed to be as successful as it is today. The fact remains the elements the android studio development environment has is perhaps what has led to its rapid growth and why most developers have embraced it. The high-end features android studio offers for java programmers are unmatched

hence the reason it is the IDE of choice when it comes to the development and testing of a mobile app.

Background

For the work of designing any mobile app, an android studio environment must be installed on the laptop and check if the system requirements for the installation are met. As postulated (Smyth) android studio is quite compatible with starting from windows 7, 8, and 10 all through to windows 11. The computer's system must have at least a minimum RAM of 8 GB in size for optimum operation during the design of the mobile apps. To get the software can easily be accessible from their website where one just downloads and follow the instructions prompt till the end part indicating to the user that it is finished.

Before proceeding into the working bench of the IDE recommendation is to configure the installation and ensure it meets the user's requirements settings. This involves specifying the user permitted to use the software although, not a must-rule. One can also decide to specify where to store their files during the development process since by default the application is normally stored in the local disk that has the operating system in its resource file folder. For successful development, another important aspect worth taking into account is to examine and ensure android studio has the expected packages installed. A user has the liberty to verify the packages installed and the ones needed if not yet installed, or others that need updating since they all work hand in hand with the IDE. Some of the most basic packages required during the development process that are necessary to install include; android SDK build tools, emulator, SDK – platform tools, google play service, and google USB drive.

Scope

The scope of this article is to give insights into the process involved in the software development of mobile apps using android studio. The concept is discussed in detail and how the android studio IDE comes into play in developing a robust app (Smyth 92). To deliver a functioning app, the problem statement to be addressed as the IDE acts as a remedy must be clearly stated and well-known from the onset. For this particular case, the problem statement is stated with the emphasis being more on comparing how developing apps now is easier for developers as opposed to before Android IDE came into existence.

Problem statement

Mobile computing as an industry before android came into play has undergone tremendous evolution dating back to the 1800s. However, since that time there has been a variety of methods used, applied techniques, API frameworks, and adopted tools when designing applications for static systems as opposed to tools for developing mobile application systems. Speaking of mobile computing systems, it is a field that has existed for quite a long but the paper's primary focus is to address the tremendous changes it has gone to date to address the challenges that have existed over time. Reflecting on history first mobile computing system to be introduced was the abacus. It was considered a mobile computing system due to its small size in nature and portable ability. Abacus is what has evolved to date to be considered a calculator. Other aspects that the abacus did not address were mainly storing and interchanging of information which came with the advent era of electronics.

Reminiscing back in the 1960s networking was introduced to allow devices to communicate by exchanging signals utilizing a wired network. Wired networks were mainly used even to date in connecting computer systems, but when it comes to mobile devices, such a

method tends to be cumbersome in the connection of mobile computing devices. During that same time, military personnel had developed alternative means of communication over the network that was mainly wireless when making a form of voice communication. Besides, the form of wireless network was also used in the sector of aviation as well as space programs apart from voice communications. They adopted the line-of-sight concept where there was no obstruction to hinder any communication wave signals. In the 1980s technological advancement led to the initiation of other technologies such as cellular telephony. The mobile phone evolution has witnessed various types of phones within the industry since then to date where people use high-end touchscreen phones that were never there before.

Furthermore, various operating systems have been used even before the establishment of the android studio which has made a lot more work easier for developers. Initially, the development of mobile applications was limited to only specific companies that produced phones that had their operating system inbuilt. Such operating systems are palm that was mainly used in windows phones while utilizing the C/C++ programming language. Another one was blackberry maintaining their proprietary Java-based platform for use precisely in blackberry devices and the final is the iPhone. All these OS limited developers to code or design an app for any suitable device because they were not open source. The mobile-producing companies had a greater advantage for enjoying the proprietary rights hence facing no competition from other device makers or developers. This was the greatest challenge in developing an application and ensuring it reaches the target audience or the market because mobile device owners had cut off such innovations get into their devices.

In that regards most mobile applications came inbuilt within the phone based on the device owners of the phone. Nevertheless, the introduction of android studio IDE came in handy

for being an open-source platform readily available for free for end users to easily download, do some alterations, and even edit. Despite the controversy surrounding open-source platforms for mobile software, android IDE has obtained traction with both device manufacturers and software developers. The platform's main developer Open Handset Alliance led by Google has made the software development more seamless and efficient. The effectiveness of the IDE comes into play due to its availability to support novice users by offering online tutorials that any interested developer can learn from their website. The platform mainly uses Java programming language and to date, most apps have been developed and deployed in play store which has the leading number of users as compared to their counterparts the iPhone.

Discussion and Analysis

Over the years' mobile telecommunication as an industry has witnessed tremendous growth with a very competitive market for mobile users. The aim of making necessary improvements to the android studio IDE is to cope with the ever-changing dynamic environment and accommodate the developer's desires or requests to ensure maximum satisfaction. The modern mobile application development styles are different from the old traditional software development tactics hence the need for one to up their software development skills to more recent. The android IDE advancement and its modern features realign well with the expectations of developing the latest version of mobile apps that suits users' needs and are readily available for downloads as opposed to before when they came already inbuilt within the phone of choice a user would choose.

The IDE provides a seamless assimilation environment that co-exists with other computing systems, for instance, databases, other web services, and legacy applications. The

methods adopted for the development of mobile applications were mainly agile development methods since it is one of the best when it comes to mobile software designing (Ye 30).

Conclusion

In conclusion, the android operating system was first developed and sponsored by Google when it came into existence across the globe many years ago, however in its tenure of culminated years in the industry, it developed a strong position and, in some respects, it beats its legendary opponents (Windows Mobile, Symbian, etc.). In this article, various aspects have been discussed related to programming for the android operating system. The introduction of the environment in which the application can be developed has been shared with the system requirements, by demonstrating the basic cornerstones of this system and a description of the functionality of some components of the android platform.

Additionally, the article has well covered the origin and the reason behind android studio as IDE became a successful platform to be used to date by most developers, The introduction states, the scene work of dedicated engineers who worked hard even in times of competition from rivals. Nevertheless, that did not deter android studio IDE to be successful and occupying a niche for itself. The paper further addresses the system requirements to be met for designing successful mobile application features when using the android platform IDE. The problem statement regarding how difficult it was for developers to design apps has been well articulated by giving the technical remedies of why the android studio has made the work of most mobile app developers easier. Finally, the paper wraps it up by giving insights into the discussion and analysis as to why the android studio is preferably still the best choice to use an IDE when working on android mobile apps.

All these discussion points towards one direction as to why the android studio was the best option to use in the designing of mobile apps precisely. The android platform has all the features intake as discussed before in integrating various components required in delivering a successful mobile app into the market. The metrics here, are to address the arising problem statement and the solution the IDE offers when it comes to what the user requires are well covered in the paper. The high-end sophisticated features that android offers to developers remain the best so far in the industry be it Java programming, C#/C++, and Kotlin which is more advanced for novice programmers but rather for experienced developers. The array of choices the android platform offers is the reason it still stands out and is the IDE of choice in delivery for this particular project.

Works Cited

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